University of Bahrain
College of Information Technology
Department of Computer Engineering

ITCE340

Introduction to MATLAB

LAB 1

Student Name:

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Introduction :

I will learn how to work with Matlab in this first lab. In order to prepare for complex code lines in the future, I will learn a few principles to become familiar with Matlab. I'll compute the DCT matrix, plot a few functions, and calculate the sum of the matrices.

Objective :

To get familiar with MATLAB.

Procedure :

1- Consider the following 8x8 input block:

A=

[450 150 350 200 125 134 32 150

250 250 150 200 123 134 32 50

130 50 250 100 120 10 32 50

50 15 250 120 123 13 32 150

220 15 250 200 120 1 4 32 20

250 10 50 40 13 134 32 40

50 10 20 20 12 13 32 30 150 10 250 200 2 0 16 32 150]

2- Calculate the sum of matrix A.

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lab1.m

```
A = [
    450, 150, 350, 200, 125, 134, 32, 150;
    250, 250, 150, 200, 123, 134, 32, 50;
    130, 50, 250, 100, 120, 10, 32, 50;
    50, 15, 250, 120, 123, 13, 32, 150;
    220, 15, 250, 200, 120, 14, 32, 20;
    250, 10, 50, 40, 13, 134, 32, 40;
    50, 10, 20, 20, 12, 13, 32, 30;
    150, 10, 250, 200, 20,16, 32,150
];

matrixA = sum(A,"all");
disp(['The sum of the matrix is: ',num2str(matrixA)]);
```

Command Window

```
The sum of the matrix is: 6730
>>
```

3- Calculate the DCT matrix A.

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lab1.m

```
A = [
    450, 150, 350, 200, 125, 134, 32, 150;
    250, 250, 150, 200, 123, 134, 32, 50;
    130, 50, 250, 100, 120, 10, 32, 50;
    50, 15, 250, 120, 123, 13, 32, 150;
    220, 15, 250, 200, 120, 14, 32, 20;
    250, 10, 50, 40, 13, 134, 32, 40;
    50, 10, 20, 20, 12, 13, 32, 30;
    150, 10, 250, 200, 20,16, 32,150
];

matrixA = dct(A);
disp(round(matrixA,2));
```

Command Window

```
  548.0100  180.3100  555.0800  381.8400  231.9300  165.4600   90.5100  226.2700
  180.3500  179.5400  158.6400   83.7000  127.6500   73.6300         0   23.7700
  137.1300   98.3200   21.3200   52.2600  -44.8900   57.3900         0   58.1400
  211.2800   15.1800  -69.1900  -24.7600  -20.4800   98.3400         0  -42.9700
   67.1800  -45.9600  222.7400  127.2800   42.4300  -40.3100         0  106.0700
  -97.1200  -74.9000  -16.4600 -115.6800  -13.5800  -39.0700         0   45.2100
  100.1000  -67.5100   79.1900  -21.6500  -19.6800   22.1500         0   29.4900
    7.1900  -36.3800   56.7900   14.1700   22.4200  -73.1600         0  -65.1500
```

4- 4- Make a plot connecting the coordinates: (2, 6), (2.5, 18), (5, 17.5), (4.2, 12.5) and (2,12) by a line.

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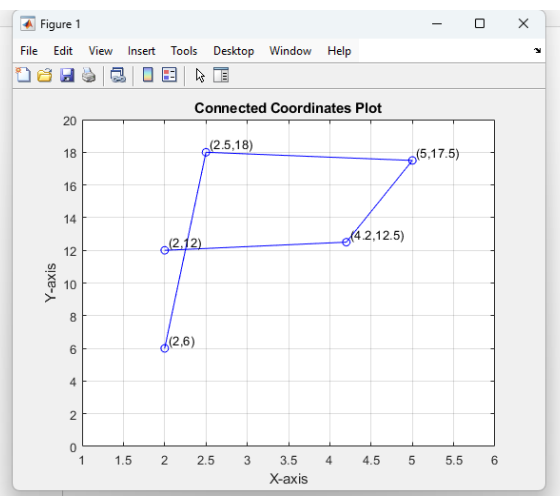
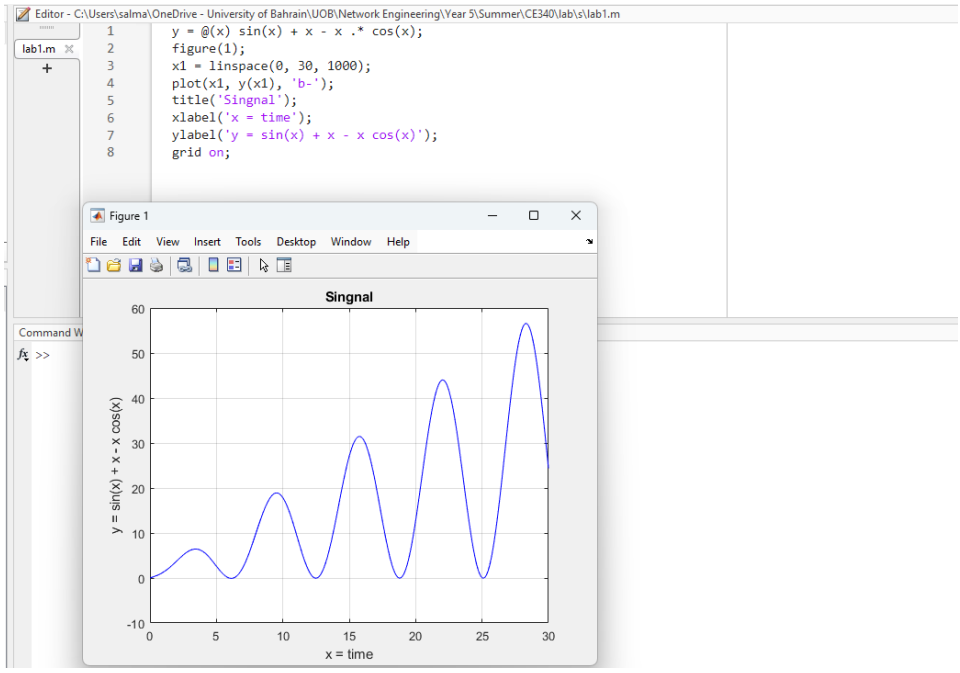
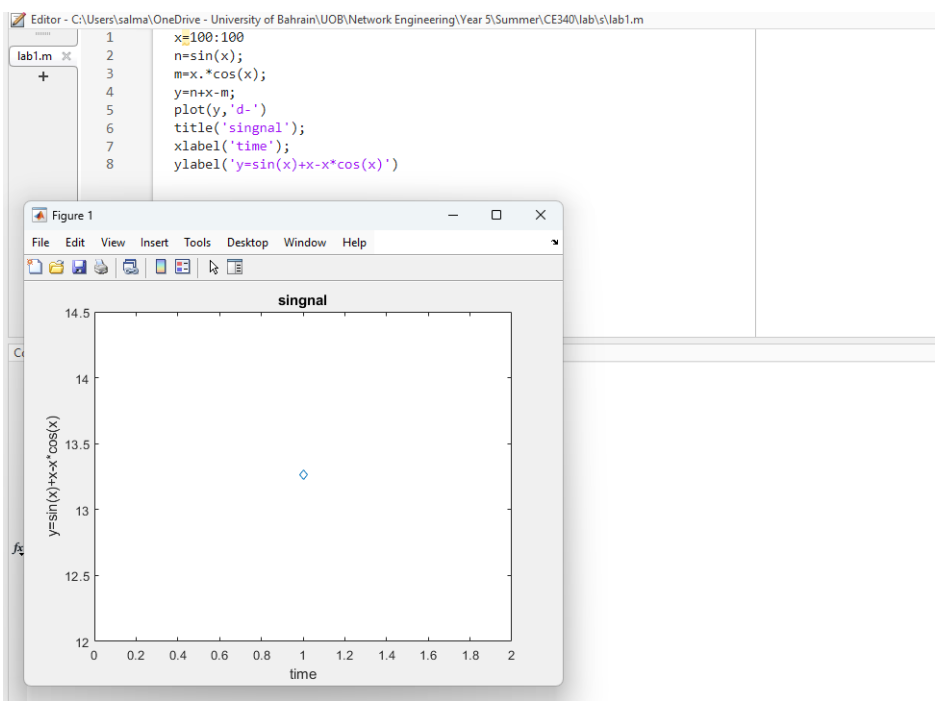
lab1.m

```
x = [2, 2.5, 5, 4.2, 2];
y = [6, 18, 17.5, 12.5, 12];
figure;
plot(x, y, 'b-o');

title('Connected Coordinates Plot');
xlabel('X-axis');
ylabel('Y-axis');
grid on;

xlim([1 6]);
ylim([0 20]);

text(x(1), y(1), ' (2,6)', 'VerticalAlignment','bottom');
text(x(2), y(2), ' (2.5,18)', 'VerticalAlignment','bottom');
text(x(3), y(3), ' (5,17.5)', 'VerticalAlignment','bottom');
text(x(4), y(4), ' (4.2,12.5)', 'VerticalAlignment','bottom');
text(x(5), y(5), ' (2,12)', 'VerticalAlignment','bottom');
```

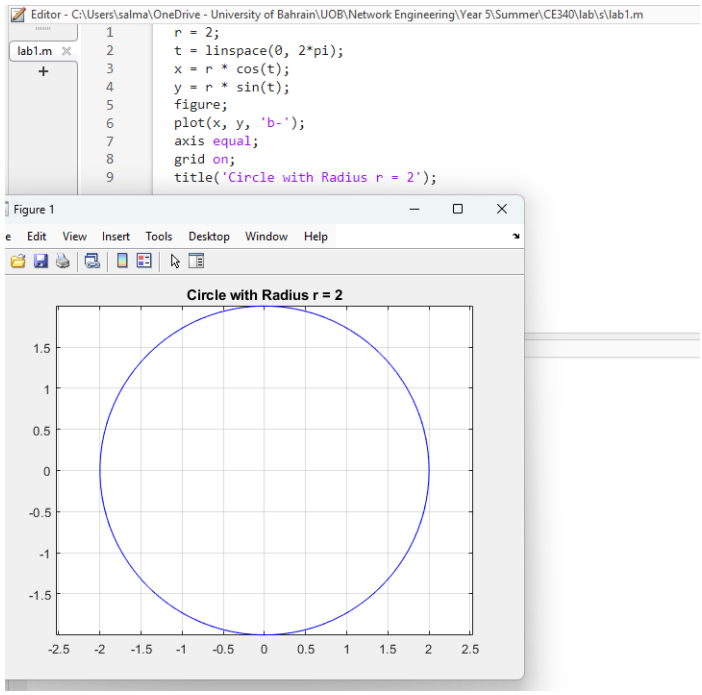


5- Plot the function y = sin(x) + x – x cos(x) in two separate figures for the intervals: 0 < x < 30 and 100 < x < 100. Add a title and axes description.



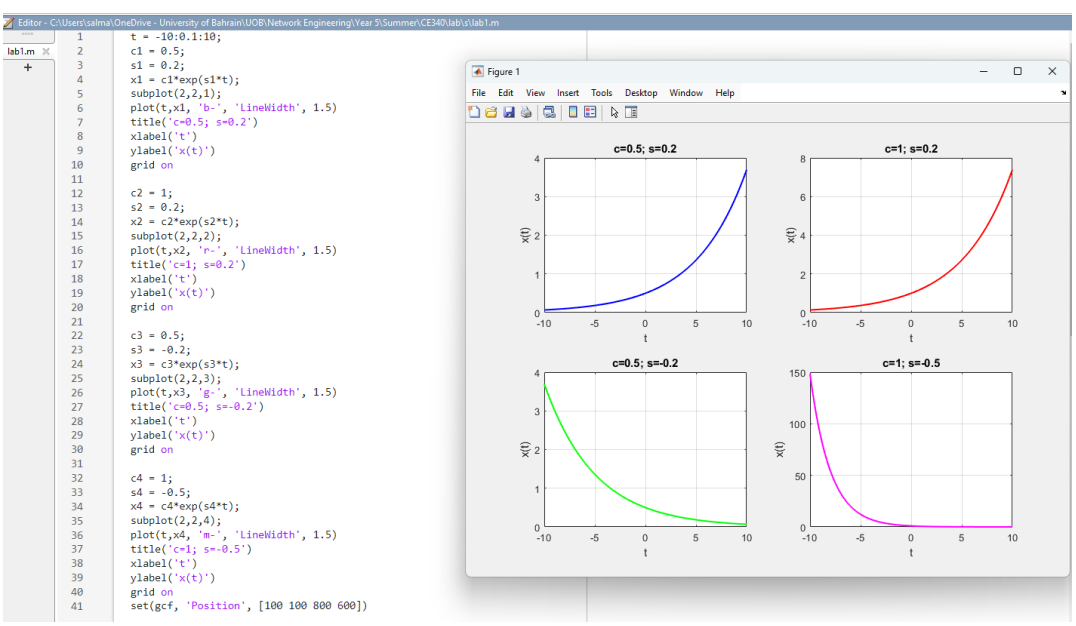


6- Plot circle with the radius r =2, knowing that the parametric equation of circle is [x(t),y(t)] =[r cos(t)]; r sin(t)] for t = [0; 2pi]

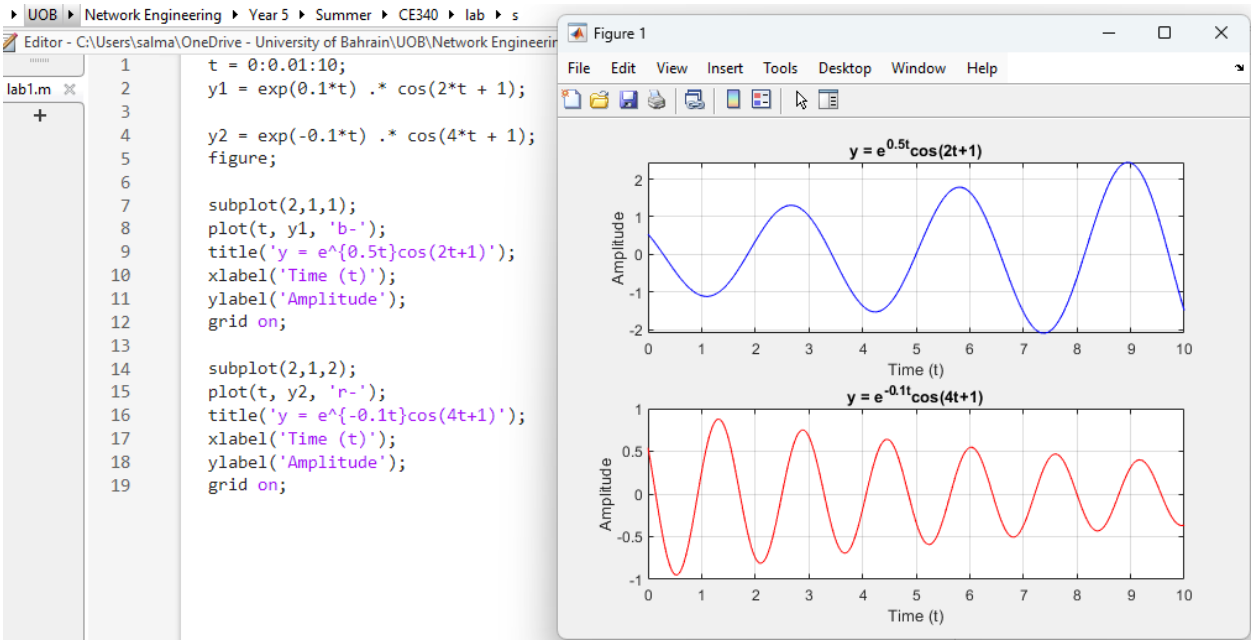


7- Plot x(t) = Ce^{at}

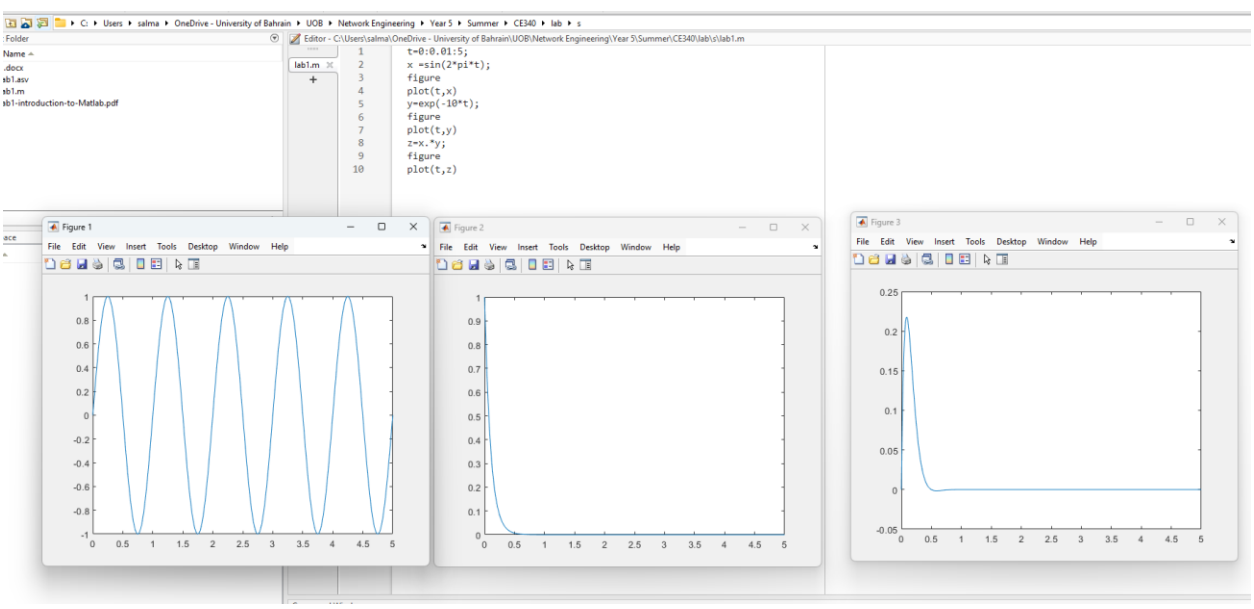
- C=0.5, a = 0.2
- C=1 , a=0.2 c- C= 0.5, a=-0.2
- C=1, a= -0.5



8- Plot : a- Exp(0.1t) cos (2t+1) b- Exp(-0.1t) cos(4t+1)



9- If x(t)=sin (2∏8t) and y(t) = e^{-5t} z= x(t) * y(t) , Plot x(t), y(t), z(t)



10- If x(t) = exp(t)

a. x(t) b. x(-t) c. exp(|t|) d. exp(-|t|)

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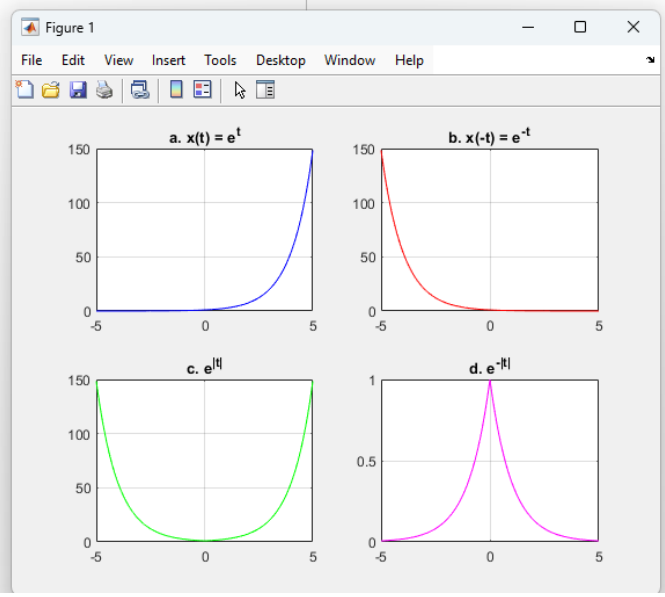
```
t = -5:0.01:5;

figure;

subplot(2,2,1);
plot(t, exp(t), 'b');
title('a. x(t) = e^t');
grid on;

subplot(2,2,2);
plot(t, exp(-t), 'r');
title('b. x(-t) = e^{-t}');
grid on;

subplot(2,2,3);
plot(t, exp(abs(t)), 'g');
title('c. e^{|t|}');
grid on;

subplot(2,2,4);
plot(t, exp(-abs(t)), 'm');
title('d. e^{-|t|}');
grid on;
```



Conclusion :

I gained knowledge of MATLAB coding at the conclusion of the first lab. I became familiar with MATLAB and learned a few concepts. Compute the matrix sum, compute the DCT matrix, then plot a few functions.